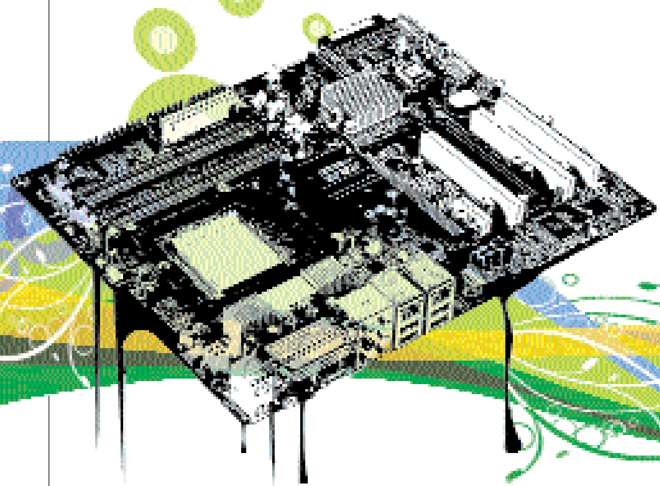

SPEEDY DEMONS

The latest dual- and quad-core desktop processors break all existing performance boundaries!

We've seen a number of improvements all designed to increase processing power. The demand for faster, more efficient computing is fuelled by the gamut of resource-hungry applications. Both industries—software and hardware—stimulate the other's growth. Of course, it is ultimately we, the users, who dictate terms to both industries, in terms of the applications we desire, and the machines we require to run them!

We've seen dual-core processors filter through to the lowest segments of our markets. This generation of CPUs delivers more performance and value than the previous generations ever did. With thermal limits being reached for the current generation processor fabrication processes, the gigahertz war has now been replaced by the quest for parallel processing. Quad-cores promise close to double the performance of today's dual-cores, and the future promises eight cores on a single CPU!

Affordability isn't an issue thanks to shrinking die sizes and optimised fabrication procedures. We've been delivered the stuff our dreams were once made off! Desktop processing has gotten a whole lot faster and economical, while keeping the environmentalists happy. Expect this trend to continue.

QUESTIONS TO ASK

Do today's processors offer a sufficient enough performance hike to substantiate an upgrade?
In a single word, Yes! The processors available today have performance gains of up to 85 per cent over previous generations. In certain cases, dual-core processors give as much as 200 per cent or more performance than older CPUs.

What are the benefits that dual-core processors will give me?
Dual-cores significantly benefit all possible PC applications and users. In general you can expect quicker response from applications especially while multi-tasking. Specific applications like gaming, video encoding and 3D rendering will benefit greatly, by as much as 60 per cent in some cases. The minimum benefits will be around 15 per cent.

Does investing in a new processor today make sense, considering that quad-cores and such are still awaited?
The upgrade path is always an uncertain one, with much promise on the horizon. Waiting for the best possible solution means waiting forever, because there's always going to be something better down the road! We recommend upgrading to a dual-core processor now if you have a processor that's more than two years old.

What about 64-bit processors?
64-bit support isn't a big deal for those not using 64-bit operating systems. However, it does add a certain amount of future proofing to your processor. When 64-bit operating systems do become *de facto* (which could be in the next two years), your CPU will be able to unleash the hidden advantages of 64-bit support.

What To Look For

A dual-core processor: They're cheap now; great value for money. Although single cores can be had for even lower prices, a single core doesn't make much sense now, since any differences in price are minimal. The sheer flexibility to multitask alone makes the multi-core processors much better options.
At least 512 KB L2 cache per core: Cache isn't always indicative of performance, but manufacturers will always give higher processor models in a particular product range a higher cache. However, comparing between different processor models using just the cache figures as a parameter will give misleading results.
Platform support: You need to select a processor based on the latest platforms. A fast CPU based on an older platform is useless simply because it's not future proof.
Clock Speed: Even though megahertz doesn't matter in CPU comparisons, it does matter when you look at CPUs belonging to the same family. For example a 3.2 GHz processor will outperform a 2.8 GHz processor from the same family of models. Note that in a particular CPU model range, the price will also increase (sometimes drastically) with increases in clock speed.

FUTURE TRENDS

Desktop processors going multi-core has brought perhaps the biggest performance hike we've seen in the last five years. We should see quad cores becoming mainstream in a couple of months. Application support isn't far behind either—Windows Vista will natively support dual cores, with strong quad core support also inbuilt. While the performance gains are still significant under Windows XP, native support means greater optimisation. A couple of games announced recently only support multi-core processors. Be warned: your single-core CPU might soon be defunct! ■